

9.1.4 Empirical Experimental Approach

This consists of repeated measurements on different test portions of a blank sample (i.e. analyte-free) or as close as possible to the supposed LOQ. This approach has the advantage of directly starting from measured data. Besides, it raises enormous practical problems, such as: how to obtain such a sample; how to deal with the qualitative but non-quantifiable data that appear around the LOQ, etc.? In conclusion of this rapid overview of the various assessments of the LOQ, it is obvious that there is no consensus.

All these procedures provide values that could be quite different. Several studies have demonstrated that LOQs obtained from identical data, but calculated according to different procedures, can be highly divergent, as explained in [2]. Moreover, it is common to obtain the value by extrapolation outside the validation domain. Indeed, in the vicinity of the LOQ, it seems logical to obtain nonquantifiable instrumental responses that cannot, by definition, be treated by classic statistical methods. Specific tools applicable to this *censored data*, mixing qualitative and quantified data, exist and are available.

This ambiguity around LOQ definitions and estimators may be surprising, considering the importance given by analysts to this parameter. It is undoubtedly imperative to establish a consensus. In any case, the recommendation to validate the LOQ rather than estimate it seems not only reasonable but mandatory. This confirmation can be done by means of surrogate samples and/or samples spiked by standard additions, then submitted to the whole analytical procedure.

All the above approaches are ultimately only estimates of the LOQ. Finally, it should be noted that in some international regulatory texts, a lower limit of quantification or LLOQ and, by symmetry, an upper limit of quantification or ULOQ are defined and required [5]. It is more a way to define the practical bounds of the validated domain rather than the whole method performance itself.

9.2 LOQ as an Expected Relative Uncertainty

The previously reported LOQ definitions and the proposed calculation procedures do not consider the measurement uncertainty (MU). Moreover, if the same experimental design is applied several times in a row, different LOQ values would be obtained. Based on this observation, another definition of LOQ that includes the MU can be proposed:

“LOQ is the lowest concentration of an analyte that can be quantified in a given matrix with a defined MU that could be absolute or relative (e.g., 50%) under specified operating conditions.”

The great interest of the uncertainty function is to predict the MU for a given measurement. When the inverse uncertainty function is taken, it can also be used to predict a concentration for a given MU value, as shown in the following formulas and explained in the Section 7.6.

Table 9.1 Different LOQ values calculated by inverting the uncertainty function from the THEOPHYLLINE study ($\mu\text{g/l}$).

	Method accuracy profile	Relative uncertainty values <i>UR</i> %				
		80%	60%	50%	30%	20%
LOQ ($\mu\text{g/l}$)	0.129	0.001	0.005	0.01	0.104	0.643

Power uncertainty function

$$UR\% = c \times X^d$$

Uncertainty function THEOPHYLLINE

$$UR\% = 0.8968 \times X^{-0.271}$$

Inverse uncertainty function

$$X = 10^{\left(\frac{\log(UR\%) - \log(c)}{d}\right)} \quad (9.1)$$

Inverse function for THEOPHYLLINE

$$X = 10^{\left(\frac{\log(UR\%) - 0.0473}{-0.271}\right)}$$

Applying this method to the results of the THEOPHYLLINE study, it is easy to construct Table 9.1, which brings together the predicted X concentrations for different values of $UR\%$. According to the new definition, each example is a possible LOQ value that can be compared to the value obtained from the accuracy profile as proposed in Section 5.2.2.

In this case, the analyst was questioned about the most *likely* LOQ value and answered for a relative uncertainty of 50% corresponding to a LOQ of 0.01 $\mu\text{g/l}$. This value is obtained thanks to an extrapolation, as the lowest experimental X -value is 0.05 $\mu\text{g/l}$. Although unacceptable for certain guidelines, it means that within the coverage interval [0.005, 0.015] $\mu\text{g/l}$ would lie the 95% proportion of possible values of a LOQ-sample. It can also be noted that the value obtained from the accuracy profile is extremely far from this proposal since 0.129 $\mu\text{g/l}$ corresponds to a relative MU of about 29% and a coverage interval of [0.092, 0.141] $\mu\text{g/l}$.

Hence, the accuracy profile (MAP)-derived approach better reflects the actual performance of the method because the LOQ is located inside the validated interval while it can be outside when derived from the uncertainty function. Another strategy is also possible, such as *a priori* defining the LOQ to be reached, computing associated relative uncertainty and coverage interval, and deciding whether it is appropriate or not.

While the MAP approach only provides an estimate of the LOQ, many regulatory bodies or official guidelines sometimes strictly require both parameters, LOD and LOQ. In this context, the question of estimating the LOD remains. Figure 9.1 is an attempt to illustrate a viable alternative solution.

With the classic approach (i.e. bottom-up), the computation starts with a blank standard deviation σ from which LOD and LOQ are deduced. As already stated, this

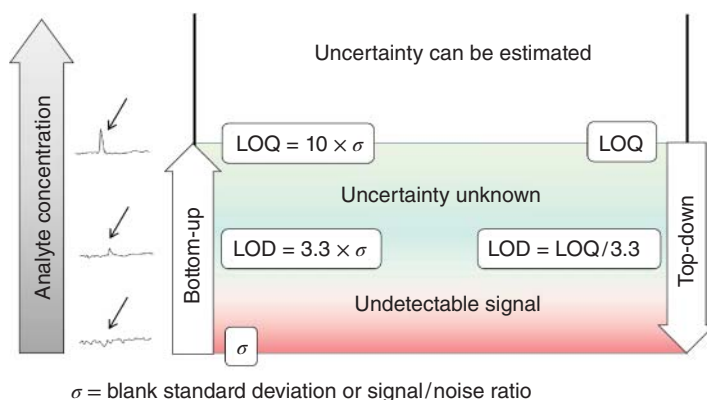


Figure 9.1 Plausible classic and alternative approaches to estimate LOD and LOQ.

parameter has some physical meaning only when working with spectrophotometric techniques. Thus, a viable alternative (i.e. top-down) is first to consider the LOQ (derived from the MAP) and then deduce the LOD. When dealing with other techniques where the signal is a peak area and/or the background noise is extremely low, such as many chromatographic methods or MS quantification, the blank standard deviation is meaningless and nonmeasurable. The estimation of LOD based on signal-to-noise ratio becomes pointless [4].

9.3 Decision Limit and Detection Capability

9.3.1 Concepts and Definitions

LOQ is often provided by equipment manufacturers or analysts to assess the performance of analytical instrumentation or method. However, according to the previous paragraphs, it is obvious that the reported values can significantly vary, as a function of the calculation procedure. This raises a question of legitimacy and applicability. If the goal is to estimate to what extent a method can quantify the lowest possible concentration, the LOQ may be the appropriate parameter.

Paradoxically, it is probably not the most interesting objective from a practical point of view since it is a question of confirming that a method is capable of quantifying concentrations in accordance with the required specifications that have been established beforehand. This goal is to verify the adequacy of the method to the use that one wants to make of it, also called fitness-for-purpose. This property of a method was introduced in the ISO 11843:2000 standard under the name Capability of Detection. The latter plays a very practical role in selecting methods able to perform the official control of foods or the environment and has been introduced in several regulatory texts.

The evolution of European regulation on food hygiene can be used to illustrate how the concept of capability of detection works and has changed over recent decades. In 1996, the European Commission published a directive “on measures

to monitor certain substances and residues thereof in live animals and animal products” [6]. But it was not until 2002 that the practical implementation of the directive was published as a Decision [7].

For the latter, the traditional statistical approach based on null hypothesis testing inspired by the LOD definition was considered. More recently, the official procedure has moved to the use of the MU and is now comparable to the procedure described in Section 8.2 about sample conformity assessment. In 2002, two parameters related to the method detection ability were introduced, namely, $CC\alpha$ and $CC\beta$, derived from the concepts developed in the ISO 11843:2000 standard [8]. In 2002, the regulatory definitions were:

<i>Decision limit</i> $CC\alpha$	“The limit at which and beyond which it is permissible to conclude with a probability of error α that a sample is noncompliant.”
<i>Detection capability</i> $CC\beta$	“Smallest level of substance that can be detected, identified, and/or quantified in a sample with a probability of error β .”

In practice, the detection capability is presented as a threshold of non-compliance, with a risk of error fixed in advance. For its calculation, the official text distinguishes between:

- The “substances for which no permitted limit has been set,” and the detection capability is comparable to the LOQ since it is the lowest concentration at which truly contaminated samples are detected.
- The “substances for which a permitted limit is set,” the detection capability is the concentration at which the method can identify samples at the set limit.

The reference in the definition to the probability of error α or β shows that it is the same logic as null hypothesis testing. The probability of error β here-mentioned must not be confused with the proportion β used to construct the β -expectation tolerance interval, abbreviated so far as β -ETI, and discussed in Chapter 5.

According to European legislation, two situations must be distinguished:

- The substance is prohibited and, therefore, must be absent.
- The substance is authorized but must not exceed a limit which can be a maximum residue limit (MRL), maximum level (ML), or “other tolerance applicable to substances” laid down in other Community legislation.

Figure 9.2 illustrates both officially defined parameters in the case of an authorized substance for which an MRL of 100 mg/kg. In this case, the required error probabilities are $\alpha = 5\%$ and $\beta = 5\%$, respectively [6].

From these probabilities, it is possible to define a coverage factor equal to 1.64 derived from the standardized normal distribution and applied to the standard deviation measured at the MRL and denoted s_{MRL} and the standard deviations applicable to a sample s_{sample} which is located “as close as possible to the MRL .”

The shortcomings of this approach are obvious to an analyst and have already been pointed out above in relation to several LOQ definitions. They can be summarized as follows:

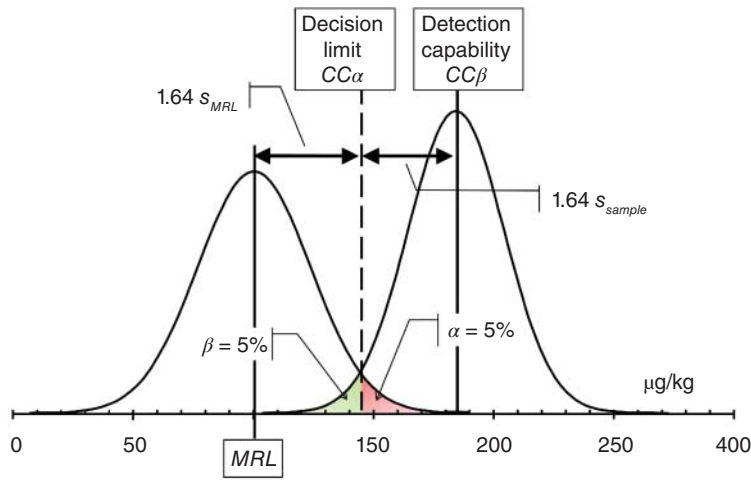


Figure 9.2 Definitions of decision limit and detection capability according EU regulation [7] for a substance with a maximum residue limit (MRL) of 100 µg/kg.

- Experimental drawbacks. How can these standard deviations be estimated? How many replicates are needed? Is it necessary to be under repeatability or intermediate precision conditions? To answer these questions globally and because these data are unknown, it is classical to use, $s_{MRL} = s_{sample}$
- Response *versus* concentration. The starting point for the proposed definitions is the instrumental response. The conversion to a concentration involves using a calibration curve, as proposed by the EPA. However, the coefficients of the curve vary from one series to another, and this variability is not considered.

For all these reasons, in 2021 the European Commission published a new regulation on “the performance of methods of analysis for residues of pharmacologically active substances used in food-producing animals and the interpretation of results and the methods to be used for sampling and repealing Decisions 2002/657/EC and 98/179/EC” [9]. In this text, the parameter names and definitions have slightly changed. In particular, new calculation methods have been introduced even if, for regulatory compatibility, the approach defined in 2002 remains applicable.

Decision limit for confirmation $CC\alpha$	“The limit at which it is safe to conclude with a probability of error α that a sample is non-compliant, with the value $1 - \alpha$ denotes the statistical certainty in percent that the allowable limit has been exceeded.”
Detection capability for screening purposes $CC\beta$	“The smallest analyte content that can be detected or quantified in a sample with a probability of error β .” In the case of prohibited or unauthorized pharmacologically active substances, the $CC\beta$ is the lowest concentration at which a method can detect or quantify, with certainty $1 - \beta$, samples containing residues of prohibited or unauthorized substances. For permitted substances, the $CC\beta$ is the concentration at which the method detects concentrations below the permitted limit with a statistical certainty of $1 - \beta$.